

Quantum vs Classical: Performance Benchmarking of CNN and QCNN in Binary Image Classification

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Abstract—This paper presents a comprehensive comparison of Convolutional Neural Network (CNN) and Quantum Convolutional Neural Network (QCNN) in cloud-based environments, focusing on training accuracy and computational efficiency. Using datasets of dog and cat images, we conducted extensive experiments on GPU platforms. Our findings indicate that QCNN consistently outperform CNN across various batch sizes and epochs, demonstrating superior accuracy and reliability in binary image classification tasks. The results highlight the potential of QCNN to revolutionize deep learning by addressing the computational limitations of classical approaches. The study also underscores the importance of quantum computing in enhancing the scalability and efficiency of neural network models.

Index Terms—QCNN, CNN, Cloud, Quantum Computing

I. INTRODUCTION

With technology evolving at a remarkable pace, we are now generating unprecedented amounts of data, much of it in the form of images. Accurately classifying and interpreting these images is essential for various applications, such as medical imaging, autonomous driving, and security surveillance. In the field of image recognition, Convolutional Neural Networks (CNN) stand out for their effectiveness in detecting and refining important image features through their systematic use of convolutional and pooling layers [1]. These layers work together to progressively refine image features, enabling the network to recognize and classify images accurately [2].

As datasets grow and models become more complex, CNNs push conventional hardware to its limits. Quantum computing, based on superposition, entanglement, and parallelism, provides far greater computational power [1]. Unlike classical bits, qubits can represent multiple states simultaneously, enabling enhanced processing efficiency [3]. In deep learning, adaptable quantum circuits that fine-tune gate parameters and perform precise measurements simplify and streamline model training.

The motivation behind this research is rooted in the groundbreaking potential of quantum computing, particularly within the sphere of quantum machine learning (QML). QCNN offer a fresh, innovative approach that uses quantum mechanics to enhance image classification. QCNN breaks through the computational barriers of traditional CNN, paving the way for advancements in fields like smart robotics, real-time analytics, and scalable data processing systems. As quantum hardware

continues to improve, developing algorithms like QCNN positions machine learning to fully tap into the powerful capabilities of quantum technology as it becomes more accessible.

In this research, our main contributions are outlined as follows:

- The study introduced a process for collecting, preprocessing, and preparing datasets for binary classification.
- To enhance efficiency, CNN and QCNN models were carefully developed and fine-tuned with a focus on batch size and epoch adjustments.
- By using established metrics, we conducted an in-depth performance assessment to compare CNN and QCNN models under a range of conditions.

We have organized the content of this paper as follows: Section II provides an overview of the existing literature, focusing on key advancements in CNN and QCNN research. Section III explores the architectural differences and design choices of CNN and QCNN models. Section IV describes the research methodology, including how data was sourced, the tools and training environments used, and the integration of hardware and software. Section V explains the evaluation metrics used to measure and validate the models' performance. Section VI delves into the experimental process, detailing the implementation of the CNN model using TensorFlow and Keras, as well as the parameter optimization. Section VII discusses the results, offering a thorough analysis of the experimental findings, such as how batch sizes and epochs impacted model performance. Lastly, Section VIII wraps up the paper by summarizing the main findings and suggesting potential directions for future research.

II. LITERATURE REVIEW

Quantum Convolutional Neural Network (QCNN) have recently emerged as an exciting alternative to classical Convolutional Neural Network (CNN) for image classification tasks. Utilizing the principles of quantum computing, QCNN aim to address some of the limitations classical CNN face, especially regarding recognition accuracy and computational efficiency. This review delves into the comparative performance of QCNN and CNN, highlighting the advancements and potential benefits found in recent studies.

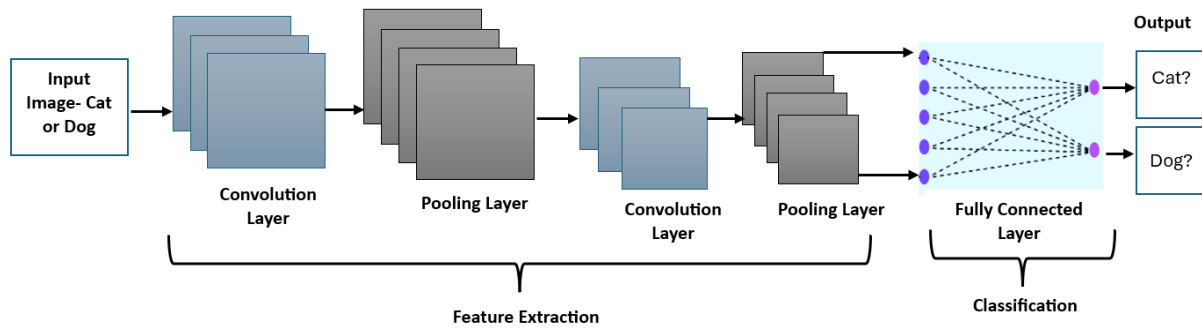


Fig. 1. Diagram of CNN Architecture.

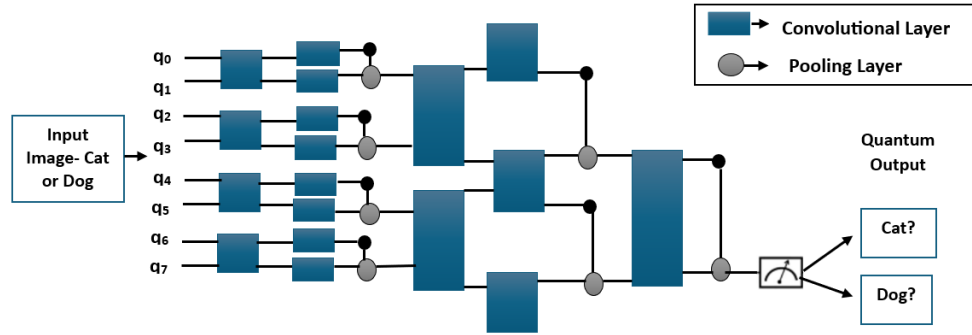


Fig. 2. Diagram of QCNN Architecture.

Recent research consistently shows that QCNN surpass classical CNN regarding image classification accuracy. For instance, Mohammed Yousif et al. (2024) introduced a novel QCNN architecture for X-ray image classification, demonstrating superior performance in both accuracy and processing time compared to classical CNN. They also conducted a comprehensive comparison, finding that QCNN achieve higher accuracy in image classification tasks [4]. These studies highlight the ability of QCNN to handle complex feature extraction more effectively than their classical counterparts. Roopa Golchha and Verma (2023) further validated these findings by comparing the performance of QCNN and classical CNN for multiclass image classification [5]. They observed that QCNN exhibit lower loss and better performance on standard datasets like MNIST and CIFAR-10, indicating that QCNN can generalize well across different types of image classification tasks. Tak Hur et al. (2021) also highlighted the capability of QCNN to outperform classical CNNs on datasets such as MNIST and Fashion MNIST, further reinforcing their effectiveness [6]. However, future advancements in quantum hardware and optimization techniques are likely to make QCNN more practical. Seunghyeok Oh et al. (2021) introduced a QRAM-based QCNN model, demonstrating its potential to lower computational demands [1].

One of the most significant advantages of QCNN is their potential for handling large-scale data. Meedinti et al. (2023) demonstrated that QCNN outperform classical CNN and ANN in accuracy and efficiency, particularly with large datasets [7].

This scalability is crucial in big data and complex machine learning applications, where traditional methods often struggle to maintain efficiency and accuracy. Varadi Rajesh et al. (2021) highlighted that QCNN can be applied in various fields like cryptography, pharmaceuticals, and signal processing, showcasing their versatility beyond just image classification tasks [8]. As highlighted by Saha et al. [9], quantum-enhanced models like quantum long-short-term memory (QLSTM) have shown significant improvements in predicting groundwater levels. Using quantum parallelism, QLSTM enhances computational efficiency and scalability, paralleling the strengths of QCNN in handling complex datasets.

III. COMPARISON OF CNN AND QCNN ARCHITECTURES

A. CNN Architecture Overview

In this study, the Fig. 1 illustrates a CNN architecture developed to classify images of cats and dogs, divided into two primary stages: feature extraction and classification. In the feature extraction stage, the input image passes through several convolutional layers that identify patterns, beginning with simple edges and shapes, and moving toward more complex structures. Each convolutional layer is followed by a pooling layer, which condenses the data, preserving essential features while reducing the spatial dimensions. After feature extraction, the data enters the classification stage, where a fully connected layer synthesizes all the gathered features to make the final prediction. This layer combines the extracted information to classify the image as either a cat or a dog.

By gradually capturing and interpreting patterns, this CNN effectively identifies the subtle differences between these two categories and provides an accurate binary classification.

B. QCNN Architecture Overview

In contrast to CNN architectures, the QCNN architecture shown in Fig. 2, connects each layer in sequence to perform hierarchical feature extraction from the input image, which is encoded across eight qubits (q0 to q7). These qubits take advantage of quantum properties like superposition and entanglement to efficiently capture complex patterns in the data. The architecture begins with convolutional layers (blue blocks), where parameterized quantum gates are applied to pairs of neighboring qubits. Each convolutional layer applies a two-qubit ansatz uniformly across all qubit pairs, allowing the network to consistently detect local features throughout the image. After each convolutional layer, a pooling layer (gray circles) reduces the number of active qubits, simplifying the data through down-sampling. This process preserves essential features and maintains consistency by using the same ansatz across the pooling layer. Each layer feeds directly into the next, enabling the network to progressively build upon previously extracted features and move from simple to more complex patterns. In the final stage, a measurement operation collapses the quantum state into a classical binary output, classifying the image as either Cat or Dog. This sequential structure allows the QCNN to distill essential features effectively, providing a potentially more efficient approach to image classification with reduced computational demands compared to classical CNN.

IV. METHODOLOGY

A. Data Acquisition

This research gathered data from two widely recognized platforms, Kaggle and Google, both known for their rich and diverse collections of datasets, including those commonly used for image classification tasks. The Kaggle dataset provided well-organized images of cats and dogs, complete with detailed annotations and labels, making it particularly useful for training CNN and QCNN. To further expand the variety of visual features, additional images representing different breeds were collected from Google, capturing a broader range of characteristics such as breed, size, and appearance. As shown in Fig. 3, the dataset includes representative images, with examples of a cat (left) and a dog (right). These datasets were meticulously prepared to support effective training, evaluation, and comparison of both CNN and QCNN models.

B. Tools, Training Environment, and Hardware Integration

We developed and tested our QCNN and CNN models in Google Colab, taking advantage of its cloud-based GPUs and the compatibility with Keras to handle large datasets efficiently [10] [11]. Although resource availability can fluctuate, we maintained consistency and reproducibility by carefully controlling parameters and configurations.

To simplify data management, we store and organize our data sets on Google Drive. For CNN models, we fine-tuned



Fig. 3. Overview of images utilized in the dataset.

GPU usage, precision settings, and data flow [12]. With the QCNN models, we refined the quantum circuits and optimized qubit utilization to enhance processing efficiency. Throughout training, we monitored various batch sizes and epochs to ensure model stability and reliability.

To evaluate the performance of CNN and QCNN models in image classification, we developed a structured workflow, as shown in Fig. 4. Starting with a dataset of dog and cat images from Kaggle, we preprocessed the images through resizing, normalization, and labeling, and then split them into training and testing sets. Both models were trained concurrently with optimized batch size and epoch settings. Finally, performance was evaluated using metrics like F1 Score, True Positive Rate (TPR), and the ROC Curve, allowing a clear comparison of each model's accuracy and efficiency.

V. EVALUATION METRICS

To understand our model's performance on new data, we relied on a set of targeted metrics designed for classification tasks. This section highlights these metrics, which help determine how well a binary classification model can distinguish between classes. Key metrics in our analysis include True Positive Rate (TPR) and False Positive Rate (FPR), which offer insights into the model's accuracy and reliability.

$$\text{True Positive Rate (TPR)} = \frac{TP}{TP + FP} \quad (1)$$

$$\text{False Positive Rate (FPR)} = \frac{FP}{FP + TN} \quad (2)$$

TPR (True Positive Rate) quantifies the proportion of correctly identified positive instances, while FPR (False Positive Rate) measures the rate of negative cases incorrectly classified as positive. True Positives (TP) represent the correctly predicted positive instances, and True Negatives (TN) denote the correctly identified negative instances. False Positives (FP) are the negative instances incorrectly classified as positive, and False Negatives (FN) are the positive instances incorrectly identified as negative. These metrics are particularly crucial in scenarios with class imbalances, offering insights into the model's ability to discriminate between classes accurately.

In addition to TPR and FPR, we also evaluated the models using Precision and Recall, which are defined as

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

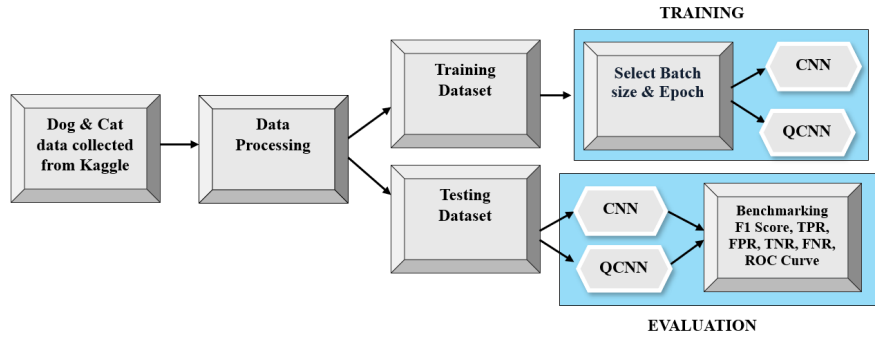


Fig. 4. CNN vs. QCNN Workflow.

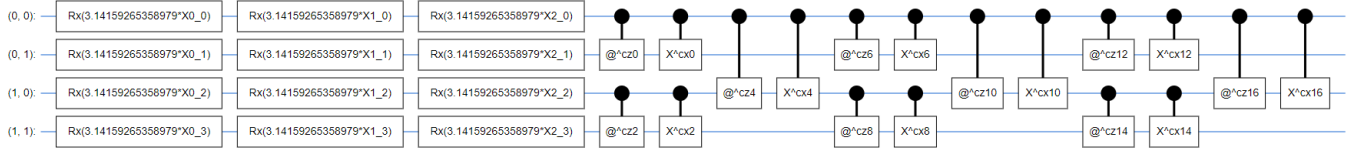


Fig. 5. QCNN Circuit Design.

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

Precision quantifies the proportion of correct positive predictions out of all positive predictions made by the model, while Recall measures the ability of the model to capture all actual positive instances. These metrics are useful for understanding the balance between correct and incorrect positive predictions.

To balance the trade-off between Precision and Recall, we employed the F1 Score, a harmonic mean of the two metrics, expressed as

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

By harmonizing precision and recall, the F1 Score provides a singular metric to estimate the model's overall performance, ranging from 0 to 1, where 1 denotes perfect precision and recall.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (6)$$

The accuracy metric provides a straightforward way to evaluate the overall performance of a classification model. It reflects the proportion of the total number of predictions (both positive and negative) that are correct.

The ROC curve and its associated AUC visually summarize how well the model balances true positives against false positives at various thresholds. By plotting the True Positive Rate (TPR) against the False Positive Rate (FPR), the ROC curve shows the model's ability to distinguish between classes. The AUC then condenses this performance into a single metric, with higher values indicating more accurate and reliable classification.

VI. EXPERIMENTAL FRAMEWORK

To classify images effectively, we implemented a CNN model using TensorFlow and Keras. Images were resized to 100x100 pixels and divided into training and validation sets. To enhance generalization, data augmentation techniques such as rotations and flips were applied, and pixel values were normalized to the [0, 1] range. The CNN architecture featured two convolutional layers with ReLU activations and max-pooling, followed by a flattening layer and two fully connected layers. The final layer used a sigmoid activation function for binary classification. Training was optimized with the Adam optimizer and binary cross-entropy loss, experimenting with batch sizes and epochs while monitoring validation loss and accuracy.

A specialized quantum convolutional layer (QConv) was developed for the QCNN using TensorFlow Quantum and Cirq to incorporate quantum computing techniques into the neural network. As illustrated in Fig. 5, the QConv layer operates on a 2x2 qubit grid at coordinates (0, 0), (0, 1), (1, 0), and (1, 1). Rx rotations parameterized by input features (X00, X01, X02, X03) are applied to each qubit. Controlled-Z (CZ) and Controlled-X (CX) gates introduce entanglement, with adjustable parameters optimized during training. A sigmoid activation function adds non-linearity, enabling the QCNN to identify complex patterns. This approach combines quantum computing's unique strengths with neural networks to enhance learning and classification performance.

VII. RESULTS AND DISCUSSION

We used an 8,000-image dataset of dogs and cats to evaluate the performance of CNN and QCNN for binary image classification. To compare the performance of CNN and QCNN, we employed various batch sizes (16, 32, and 64) and epochs

TABLE I
CONFUSION MATRIX FOR CNN AND QCNN MODELS

	Predicted Positive		Predicted Negative	
	<i>CNN</i>	<i>QCNN</i>	<i>CNN</i>	<i>QCNN</i>
Actual Positive	387	458	113	42
Actual Negative	85	14	415	486

TABLE II
PERFORMANCE COMPARISON OF CNN AND QCNN

Metric	<i>CNN</i>	<i>QCNN</i>
F1 Score	0.79	0.94
Accuracy	80%	94.4%

(5, 10, 15, and 20). This setup facilitated analysis, allowing us to test the models using a smaller, 1,000-image dataset.

Table I shows that after 10 epochs with a batch size of 16, the QCNN notably outperforms the CNN. QCNN correctly classifies 458 positives and 486 negatives, with just 42 false negatives and 14 false positives. In contrast, the CNN misclassifies more cases, proving less reliable for high-precision tasks. Further tuning, such as refining thresholds and adding layers, only strengthened QCNN's performance, underscoring its potential in real-world scenarios.

Table II provides a side-by-side performance comparison of CNN and QCNN models, evaluated at epoch 10 with a batch size of 16, using F1 Score and Accuracy as benchmarks. The QCNN clearly takes the lead, demonstrating a substantial edge in both metrics. It achieves an impressive F1 Score of 0.94, well above the CNN's 0.79, and an accuracy of 94.4%, compared to the CNN's 80%. These results emphasize QCNN's exceptional capability in managing class imbalances, positioning it as an ideal choice for tasks that require both precision and robustness.

A comparison of the number of TP the CNN and the QCNN achieved across three different batch sizes: 16, 32, and 64, is shown in Fig. 6. The results indicate that the QCNN consistently outperforms the CNN regarding true positive counts at each batch size. This performance difference illustrates the potential of quantum computing in enhancing model accuracy. The figure also highlights the robustness of

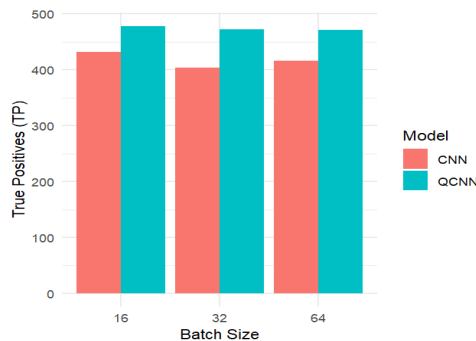


Fig. 6. TP for CNN and QCNN across different batch sizes.

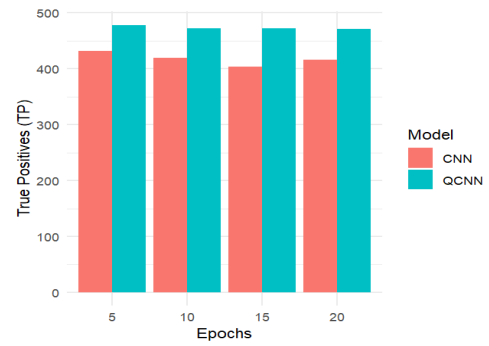


Fig. 7. TP for CNN and QCNN across different epochs.

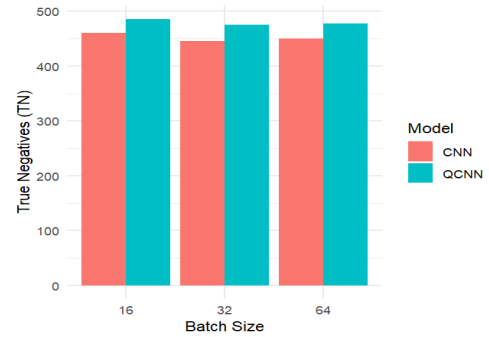


Fig. 8. TN for CNN and QCNN across different batch sizes.

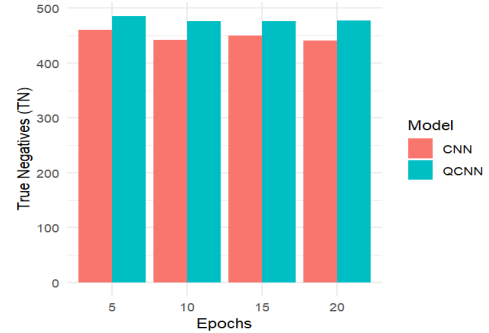


Fig. 9. TN for CNN and QCNN across different epochs.

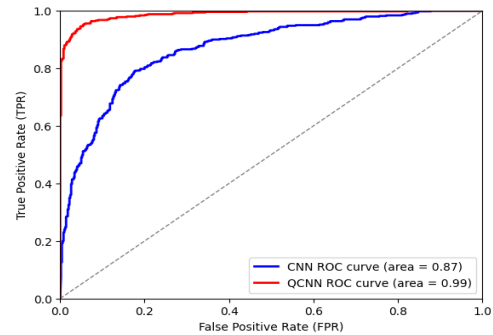


Fig. 10. ROC Curves Comparing CNN and QCNN Models.

the QCNN model across various batch sizes, suggesting its effectiveness in different training configurations.

The performance of CNN versus QCNN in terms of TP over various epochs: 5, 10, 15, and 20 is illustrated in Fig. 7. QCNN consistently achieve higher TP counts compared to CNN at each epoch, underscoring the enhanced accuracy of QCNN in identifying true positives. This trend suggests that QCNN maintain superior detection capabilities throughout the training process, offering potential improvements in model performance and reliability over extended training periods.

The bar chart illustrates the performance of CNN and QCNN models in terms of True Negatives (TN) across different batch sizes (16, 32, and 64). The QCNN model consistently achieves more True Negatives than the CNN model at all batch sizes, as shown in Fig. 8. Specifically, at a batch size of 16, the QCNN model surpasses the CNN model by a significant margin. This trend continues at batch sizes 32 and 64, where the QCNN model maintains its superiority in detecting True Negatives. This indicates that the QCNN model is more effective in correctly identifying negative cases regardless of the batch size.

The chart highlights the comparison of TN between CNN and QCNN models over different numbers of epochs (5, 10, 15, and 20) in Fig. 9. Across all epoch intervals, the QCNN model consistently outperforms the CNN model in correctly identifying negative cases. Notably, the QCNN model achieves a higher number of True Negatives at each epoch checkpoint, with the gap between the two models being most evident at the 5-epoch mark. This consistent performance of the QCNN model across varying epochs suggests its robustness and superior ability to maintain higher accuracy in negative case identification over extended training periods.

The ROC curves illustrate the performance comparison between CNN and QCNN models regarding their TPR against the FPR, as shown in Fig. 10. The QCNN model performs significantly better, with an area under the curve (AUC) of 0.99, almost achieving perfect accuracy. This means the QCNN model correctly identifies positive cases while minimizing false positives. On the other hand, the CNN model has an AUC of 0.87, indicating it is less accurate in classification. The distinct separation between the two curves emphasizes the QCNN model's enhanced ability to discriminate between positive and negative cases compared to the CNN model. This highlights the QCNN model's effectiveness and robustness in classification tasks.

VIII. CONCLUSION AND FUTURE RESEARCH

This study highlights the transformative potential of QCNN over classical CNN, proving that QCNN consistently deliver higher accuracy and efficiency in binary image classification. By tapping into quantum computing's power, QCNN show promise in tackling the limitations faced by traditional approaches, especially with complex datasets and large-scale applications. Our results suggest that QCNN could reshape the future of deep learning, offering a scalable and highly

accurate alternative that's well-suited for real-world scenarios demanding precision.

Looking ahead, there is much room to grow. Fine-tuning quantum circuits, testing on larger datasets, and creating hybrid models that blend quantum and classical approaches could unlock even greater potential. Beyond binary classification, QCNN could make a real impact in areas like medical imaging, drug discovery, and natural language processing. As they expand into multiclass classification, QCNN stand ready to shape the future of AI, offering powerful new tools for some of the field's most difficult problems. The journey has just begun, and the possibilities are truly exciting.

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